

c.c(ou)

16/06/21

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$$\begin{array}{r} 36038 \\ -99 \times \equiv_{19} 44 \end{array} \quad \begin{array}{r} 5491 \\ 44 \end{array}$$

SVLG

Riduzione BASI!

1. STEP, RIDUZIONE BASI.

$$99 = 19 \cdot (5) + 4$$

$$44 = 19 \cdot 2 + 6$$

$$99:19 = 5$$

$$5 \times 19 = 95 \text{ NE MANCANO } 4$$

Sostituisco:

$$\begin{array}{r} 36038 \\ -4 \end{array} \equiv_{19} \begin{array}{r} 5491 \\ 6 \end{array}$$

2. STEP, TROVARE GLI MCD.

$$\text{MCD}(4, 19) = 1$$

$$\text{MCD}(6, 19) = 1$$

3. TEOREMA DI EULERO

$$4^{\phi(19)} \equiv_{19} 1 \quad 1^{\text{a}} \text{ BASE}$$

$$6^{\phi(19)} \equiv_{19} 1 \quad 2^{\text{a}} \text{ BASE}$$

$$\phi(19) = 18 \rightarrow \text{ho ridotto la congruenza}$$

4. RIDURRE GLI ESPONENTI

$$36038 = 18 \cdot 2002 + 2$$

$$5491 = 18 \cdot 305 + 1$$

5. UNIONE/ACCROCCO

$$\begin{aligned} 36038 &\equiv_{19} \left(4^{\phi(19)} \right)^{2002} \cdot 4^2 \equiv_{19} 4^2 \equiv_{19} 16 \equiv_{19} -3 \\ 16 &= 19 \cdot 1 - 3 \end{aligned}$$

$$5491 \equiv_{19} \left(6^{\phi(19)} \right)^{305} \cdot 6 \equiv_{19} 6$$

6. SOSTITUZIONE finale

$$-(-3) \times \equiv_{19} 6$$

$$3 \times \equiv_{19} 6$$

$$x \equiv_{19} 2$$

$$S = \{ z + 19z \mid z \in \mathbb{Z} \}$$

